Response to Reviewers' Comments

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Dear Editor and Reviewers,

Thank you for your time and constructive comments on our manuscript. We provide a point-by-point response below, explaining the changes made and identifying their locations in the revised manuscript.

All revised passages are highlighted in blue in the marked version of the revised manuscript, and the page and line numbers in this response refer to that version. Reference numbers refer to the reference list of the revised manuscript, and equation numbers to the original manuscript unless stated otherwise.

The main revisions are summarized as follows:

1. *Threat model.* The per-AP separation of the adversary is now stated as a genie-aided strong-adversary assumption, and the identifiability condition of the per-AP reconstruction is given (Reviewer 1, Comment 1; Reviewer 2, Comment 2).

2. *Sensing model.* The receive processing, the phase-averaged sensing SNR, the treatment of the AN echo, and the detector are now specified, and the sensing-SNR loss is translated into detection probabilities (Reviewer 1, Comment 4; Reviewer 2, Comment 1; Reviewer 3, Comments 1, 2, and 5).

3. *Fake-peak constraint.* The constraint now includes the true-direction AN response, so that $\rho > 1$ guarantees that the fake-direction response exceeds the true-direction one, and the virtual location is chosen such that all transmit APs resolve the two directions (Reviewer 1, Comment 2).

4. *Privacy evaluation and claims.* A multi-hypothesis adversary is evaluated, the property that governs it is identified, the claims are narrowed to what is guaranteed, and the estimated locations are visualized (Reviewer 1, Comment 3; Reviewer 2, Comment 5).

5. *Baselines and tradeoffs.* A robust no-AN baseline, a no-AN alternative that spends the AN power on communication, and a more favorable adversary are evaluated (Reviewer 1, Comment 5; Reviewer 3, Comments 3 and 4).

6. *Synchronization.* The effect of residual inter-AP phase offsets on the effective-channel null is analyzed in closed form and evaluated numerically (Reviewer 2, Comment 4).

7. *Literature and minor corrections* (Reviewer 2, Comment 3 and minor corrections; Reviewer 3, Comment 5).

All numerical results have been recomputed with the revised design.

Thank you for considering our revised manuscript.

Yours sincerely,

[Corresponding author's name]
On behalf of all authors

Comments by Reviewer 1

Comment 1:

The adversary is assumed to reconstruct the transmit signals of individual APs from its received observations. Since the received signal contains a superposition of multiple AP transmissions, it is unclear how these components can be reliably separated, especially in the presence of AN and imperfect CSI.

Response:

We thank the reviewer for this important comment and agree that the separation step requires an explicit treatment. Since all APs transmit common symbols, the aggregate observation $\mathbf{y}_a[n] = \sum_k \mathbf{H}_{a,k}\mathbf{x}_k[n] + \mathbf{n}_a[n]$ provides only M_{UE} observation dimensions per snapshot for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions. Blind separation of the per-AP components from $\mathbf{y}_a[n]$ alone is therefore not possible in general, and we do not claim otherwise. The original manuscript did not state clearly that this step is an assumption, and we have revised the threat model accordingly.

Following the threat model of [14], the revised manuscript explicitly adopts a *genie-aided strong-adversary model*, in which the per-AP components of the received samples are made available to the adversary. We adopt this model deliberately, for the following reasons.

1. *Conservative privacy assessment.* The genie removes the inter-AP superposition, which is the main obstacle for a practical adversary seeking AP-dependent directions. The privacy of the proposed design is therefore assessed against an adversary that is granted a capability that a practical adversary does not have in general.
2. *Plausibility.* Partial per-AP separation may become available in practice, e.g., when AP-specific transmissions are resolvable in time or frequency, or when the propagation delays of the distributed APs are resolvable over a wide bandwidth.
3. *Design principle.* The network cannot verify the receiver capability of a served user. A privacy mechanism should therefore not rely on the adversary being unable to separate the per-AP signals.

Regarding the AN, it is intentionally contained in the per-AP components available to the adversary. Keeping the AN visible in the raw observations, while nulling it at the legitimate combiner outputs, is the receive-processing asymmetry on which the proposed design relies. Regarding imperfect CSI, the adversary's channel-estimation error enters the residual term $\widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ of the signal model. Consistent with the strong-adversary model, the simulated adversary is further given accurate knowledge of its per-AP channels $\mathbf{H}_{a,k}$, which is now stated in Section IV. The identifiability of the per-AP reconstruction step is discussed in our response to Reviewer 2, Comment 2.

Location in the revised manuscript: Section II, threat-model paragraph; Section IV-A, simulation setup; Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

Section II: "Since all APs transmit common symbols, $\mathbf{y}_a[n]$ offers only M_{UE} dimensions for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions, so its per-AP decomposition is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [14], we therefore adopt a genie-aided strong-adversary model in which the per-AP components of the received samples are available to the adversary, as may occur when AP-specific transmissions are resolvable in time, frequency, or delay. Since the network cannot verify the receiver capability of a served user, this conservative model avoids relying on the adversary being unable to separate them."

Section IV-A: "Throughout, the beams are designed from LMMSE channel estimates, whereas the realized SINRs are evaluated on the actual channels, and the genie-aided adversary of Section II is further given accurate knowledge of $\mathbf{H}_{a,k}$."

Section IV-B: "Since the adversary is genie-aided, these success rates represent the leakage under a conservative attack model."

Comment 2:

Constraint (9g) does not necessarily guarantee that the fake direction becomes the dominant peak of the reconstructed beampattern. Moreover, the condition used to derive (10) is not explicitly enforced in the optimization.

Response:

We thank the reviewer for this careful observation and agree with both points. Under the ideal covariance decomposition, the reconstructed true-direction response is $\widehat{B}_k(\phi_k) = S_k + T_k$, where $S_k = \|\mathbf{a}^H(\phi_k)\mathbf{W}_{-a,k}\|_F^2$ and $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. The original constraint (9g) bounds S_k through the fake-direction response $F_k = |\mathbf{a}^H(\phi_{\mathrm{fake},k})\mathbf{z}_k|^2$, but places no restriction on T_k, i.e., on the AN response in the true direction. The fake-direction response could therefore fall below the true-direction response even when (9g) holds, and the condition $T_k/F_k \le 1/M_{\mathrm{tx}}$ used to derive (10) was indeed not enforced. We have revised the design as follows.

(i) Revised fake-peak constraint. We now include the true-direction AN response in the constraint itself:

$$\left|\mathbf{a}^H(\phi_{\mathrm{fake},k})\mathbf{z}_k\right|^2 \ge \rho \left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2 + \left|\mathbf{a}^H(\phi_k)\mathbf{z}_k\right|^2, \quad k \in \mathcal{K}_{\mathrm{res}}. \tag{9g$'$}$$

Since the remaining terms of $\widehat{B}_k(\phi_{\mathrm{fake},k})$ are nonnegative, $\widehat{B}_k(\phi_{\mathrm{fake},k}) \ge F_k$, and since $S_k \le \|\mathbf{a}^H(\phi_k)\mathbf{W}_k\|_F^2$, (9g$'$) directly gives

$$\widehat{B}_k(\phi_{\mathrm{fake},k}) - \widehat{B}_k(\phi_k) \ge F_k - S_k - T_k \ge (\rho - 1)\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2. \tag{10$'$}$$

Hence, $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response, and this guarantee now follows from a constraint that the optimization enforces, without any auxiliary condition. The previous sufficient condition $\rho > M_{\mathrm{tx}}/(M_{\mathrm{tx}} - 1)$ is replaced by $\rho > 1$.

(ii) Tractability. In the CCCP update, (9g$'$) becomes $\rho\,\mathrm{Tr}(\mathbf{W}_k^H\mathbf{A}_k\mathbf{W}_k) + \mathbf{z}_k^H\mathbf{A}_k\mathbf{z}_k - \widetilde{\zeta}_k \le 0$, where $\widetilde{\zeta}_k$ is the affine lower bound of F_k. This is a convex quadratic constraint. Since $\widetilde{\zeta}_k \le F_k$ globally, every point feasible for the surrogate is feasible for (9g$'$), so the feasibility, monotonicity, and convergence arguments of Section III-B carry over unchanged.

(iii) Resolvability. The comparison in (10$'$) is meaningful only when the M_{tx}-element array can resolve the true and fake directions. We now make this explicit by imposing (9g$'$) on

$\mathcal{K}_{\text{res}} = \{k : |\sin\phi_k - \sin\phi_{\text{fake},k}| \geq 2/M_{\text{tx}}\}$, i.e., on the APs whose two directions are at least one Rayleigh width apart. With the original virtual location $\mathbf{p}_{\text{f}} = (5, -25)$ m, the target and the virtual location were nearly collinear with three transmit APs, for which $|\sin\phi_k - \sin\phi_{\text{fake},k}| \approx 0.03$–$0.04 < 2/M_{\text{tx}} = 0.125$, so no distinguishable fake peak could be formed at these APs. We have therefore relocated the virtual location to $\mathbf{p}_{\text{f}} = (175, 35)$ m, for which all eight transmit APs satisfy the condition, with $\min_k |\sin\phi_k - \sin\phi_{\text{fake},k}| = 0.163$, so that $\mathcal{K}_{\text{res}}$ contains all transmit APs. As explained in our response to Reviewer 2, Comment 5, this full support is also what makes the true and fake hypotheses symmetric for a multi-hypothesis adversary. The set $\mathcal{K}_{\text{res}}$ is kept in the formulation, since it determines where the constraint is meaningful in a general geometry.

(iv) From the true direction to the dominant peak. We agree that (10′) compares the fake direction with the true direction only, and that it holds for the ideal covariance. It does not guarantee that $\phi_{\text{fake},k}$ is the global maximum over all θ, since the beams of the other users produce responses in their own directions, and the finite number of snapshots and the EM reconstruction error add residual cross terms. We have therefore revised the manuscript to claim only what is guaranteed, namely that the fake-direction response exceeds the true-direction response, and we evaluate the location of the reconstructed dominant peak numerically. For localization deception, the relevant event is that the dominant peak is displaced from the true direction, since a peak elsewhere also removes that AP's direction line from the target cell. At $\eta \approx 0.50$, over the APs in $\mathcal{K}_{\text{res}}$, the reconstructed fake-direction response exceeds the true-direction response in $[X]$% of the cases, and the dominant peak lies at the fake direction, the true direction, and elsewhere in $[X]$%, $[X]$%, and $[X]$% of the cases, respectively.

(v) Updated results. All results have been recomputed with (9g′). At $\eta \approx [X]$, the proposed design now reduces the localization success rate to $[X]$%, with a sensing SNR of $[X]$ dB.

Location in the revised manuscript: Section III-B, problem (9) and the paragraph containing (10); the CCCP update (12); Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

Section III-B, constraint (9g):

$$\left|\mathbf{a}^H(\phi_{\text{fake},k})\mathbf{z}_k\right|^2 \geq \rho \left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2 + \left|\mathbf{a}^H(\phi_k)\mathbf{z}_k\right|^2, \ \forall k \in \mathcal{K}_{\text{res}}.$$

Section III-B, paragraph containing (10): "With $\mathbf{W}_{-a,k}$ denoting $\mathbf{W}_k$ without $\mathbf{w}_{a,k}$, define $S_k = \|\mathbf{a}^H(\phi_k)\mathbf{W}_{-a,k}\|_F^2$, $F_k = |\mathbf{a}^H(\phi_{\text{fake},k})\mathbf{z}_k|^2$, and $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. The ideal covariance decomposition gives $\widehat{B}_k(\phi_{\text{fake},k}) \geq F_k$ and $\widehat{B}_k(\phi_k) = S_k + T_k$, so (9g) yields

$$\widehat{B}_k(\phi_{\text{fake},k}) - \widehat{B}_k(\phi_k) \geq F_k - S_k - T_k \geq (\rho - 1)\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2.$$

Hence, $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response. Since this comparison requires resolvable directions, (9g) is imposed on $\mathcal{K}_{\text{res}} = \{k : |\sin\phi_k - \sin\phi_{\text{fake},k}| \geq 2/M_{\text{tx}}\}$. The guarantee holds for the ideal covariance and relative to the true direction; the location of the reconstructed dominant peak under finite snapshots and reconstruction errors is evaluated in Section IV."

Section III-B, CCCP update (12), last constraint: "$\rho\,\text{Tr}(\mathbf{W}_k^H\mathbf{A}_k\mathbf{W}_k) + \mathbf{z}_k^H\mathbf{A}_k\mathbf{z}_k - \widetilde{\zeta}_k \leq 0, \ \forall k \in \mathcal{K}_{\text{res}}$."

Section IV-B: "By (10), $\rho > 1$ is sufficient for the fake-direction response to exceed the true-direction response at every transmit AP, since $\mathbf{p}_{\mathrm{f}}$ is chosen such that $\mathcal{K}_{\mathrm{res}}$ contains all of them."

Table I: "True/fake coordinates: $(-75, 75)/(175, 35)$ m."

Comment 3:

The privacy evaluation considers only an adversary that estimates the target location from the dominant beampattern directions. It would be useful to examine whether the proposed method remains effective against more sophisticated localization strategies. Also, a two-dimensional visualization of the estimated target locations would help demonstrate whether the estimates are actually shifted toward the intended fake location.

Response:

We thank the reviewer for these two suggestions, both of which we have followed.

More sophisticated localization strategies. In addition to the dominant-peak attack of [14], the revised manuscript evaluates a multi-hypothesis adversary. It retains the two strongest distinct peaks of each reconstructed beampattern $\widehat{B}_k(\theta)$, casts one direction line per peak, and forms the two most-voted, geometrically separated location hypotheses. It then resolves them by the vote count, at random, or, with knowledge of the fake-peak mechanism, by choosing the hypothesis supported by the weaker peaks. We also report the success of an adversary allowed two guesses, which bounds every rule that picks one of the two hypotheses. At $\eta \approx 0.50$, the success rates of the three rules are $[X]\%$, $[X]\%$, and $[X]\%$, and the two-guess success is $[X]\%$, compared with $[X]\%$, $[X]\%$, $[X]\%$, and $[X]\%$ without AN. As discussed in detail in our response to Reviewer 2, Comment 5, the AN cannot remove the true-direction peak, which the legitimate sensing requires. The proposed design therefore creates a competing hypothesis rather than erasing the true one, and the parameter ρ trades the deception of the dominant-peak attack against the indistinguishability of the two hypotheses for a mechanism-aware adversary.

Two-dimensional visualization. Fig. R1 shows the locations estimated by the dominant-peak attack over all channel realizations, without AN and with the proposed design at $\eta \approx 0.50$. Without AN, the estimates concentrate at the target, with $[X]\%$ of them in the target cell. With the proposed design, they concentrate at the virtual location, with $[X]\%$ of them in its cell. The estimates are therefore drawn toward the intended fake location rather than merely scattered away from the target. Given the length limit of a correspondence, Fig. R1 is provided in this response, and Section IV-B reports the two fractions.

[Fig. R1: generated by `fig.7.detection/plot_location_scatter.m`, file `location_scatter_response.pdf`.]

Fig. R1. Locations estimated by the dominant-peak attack without AN (left) and with the proposed design at $\eta \approx 0.50$ (right). Marker area is proportional to the number of channel realizations; $\times$: target, $+$: virtual location, $\triangle/\square$: transmit/receive APs.

Location in the revised manuscript: Section II, threat model; Section IV-B, discussion of Fig. 2. [Add page and line numbers.]

Revised text:

> *Section II:* "Besides this dominant-peak attack, we consider a multi-hypothesis adversary that retains the two strongest distinct peaks of each $\widehat{B}_k(\theta)$, forms the two most-voted, geometrically separated location hypotheses, and resolves them by the vote count, at random, or with knowledge of the fake-peak mechanism."
>
> *Section IV-B:* "Without AN, $[X]$% of the estimates fall in the target cell, whereas with the proposed design at $\eta \approx 0.50$, $[X]$% fall in the cell of $\mathbf{p}_{\mathrm{f}}$, so the fused estimate is drawn toward the virtual location rather than merely perturbed. Against the multi-hypothesis adversary, the success rates at $\eta \approx 0.50$ are $[X]$%, $[X]$%, and $[X]$% for the vote, random, and mechanism-aware rules."

Comment 4:

The sensing SNR expression in (2) needs further clarification. The contributions from different APs are added without considering possible coherent terms. In addition, the AN waveform is assumed to be known at the legitimate sensing receivers, but its contribution is excluded from the sensing SNR.

Response:

We thank the reviewer for this comment. We have revised the sensing model in Section II to state the receiver processing and the assumptions behind (2) explicitly, and we address the two points below.

Coherent cross terms. Since all transmit APs radiate common symbols, the echo received at receive AP r is the superposition $\sum_k \alpha_{r,k}\sqrt{g_{r,k}}\,\mathbf{a}^H(\phi_k)\mathbf{W}_k\mathbf{s}[n]$, where $\mathbf{s}[n]$ collects the symbols and $\alpha_{r,k}$ is the complex bistatic echo coefficient with $\mathbb{E}\{|\alpha_{r,k}|^2\} = \xi_{r,k}^2$. Its instantaneous power therefore contains cross terms between transmit APs, whose values depend on the relative phases of $\{\alpha_{r,k}\}_k$. These phases are not known to the network, because the target position is known only coarsely relative to the carrier wavelength ($\lambda_c \approx 15.8$ cm at 1.9 GHz) and the target response varies with the aspect angle. We therefore model them as independent and uniformly distributed across transmit–receive AP pairs. Under this model, the cross terms vanish in expectation, and (2) is the per-snapshot sensing SNR averaged over the echo phases, which is why it is referred to as the *average* sensing SNR. The revised manuscript now states this assumption explicitly. A design that exploits coherent superposition at the target would require phase-accurate knowledge of the target position and phase synchronization of the transmit APs at the target, which we do not assume for the sensing function.

AN contribution. The reviewer is correct that the AN echo was excluded from (2) without explanation. Since $s_z[n]$ is known at the receive APs and independent of the data and sensing symbols, its echo is separated by matched filtering and could in principle be exploited. We keep it out of the design objective deliberately. The AN echo from the true target is proportional to $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$, so including it in the objective would reward AN power in the true direction. This would raise the true-direction peak of the beampattern reconstructed by the adversary and work against the fake-peak constraint. Exploiting the AN echo at the receivers after the design would add $\sum_k c_k T_k \geq 0$ to γ_t. Hence, the reported γ_t is a lower bound on the achievable sensing SNR, and the reported 3.6 dB loss is an upper bound on the sensing cost of the proposed design. Both points are now stated in Sections II and III.

Location in the revised manuscript: Section II, sensing-model paragraph; Section III-B, paragraph following (10). [Add page and line numbers.]

Revised text:

> *Section II:* "Each receive AP employs M_{rx} antennas, steers its receive beam toward the target, and matched-filters its received samples with the known transmitted symbols, where the data symbols, being available at the central processor, act as additional probing signals. The bistatic echo phases are not known to the network to within the carrier wavelength and vary with the aspect angle, and are therefore modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases removes the cross terms between the echoes of different transmit APs. With $\mathbf{W}_k \triangleq [\mathbf{w}_{1,k}, \ldots, \mathbf{w}_{N_{\mathrm{UE}},k}, \mathbf{w}_{t,k}]$, the per-snapshot sensing SNR, averaged over the echo phases and the N_{rx} receive APs, is [(2), unchanged]."
>
> *Section III-B:* "Since $s_z[n]$ is known at the receive APs and independent of the other symbols, matched filtering separates its echo. This echo is excluded from (2) and from the design objective, so that the objective does not reward AN power in the true direction, which would raise T_k. Exploiting it at the receivers would add $\sum_k c_k T_k \geq 0$, so γ_t is a lower bound on the achievable sensing SNR."

Comment 5:

The simulation results could be strengthened. In particular, Fig. 4 compares the proposed robust design mainly with nominal schemes. Including a robust no-AN baseline would provide a fairer assessment of the benefit of the proposed approach.

Response:

We thank the reviewer for this suggestion and agree that the comparison in Fig. 4 was not balanced. We have added a robust no-AN baseline, obtained by applying the same error-averaged SINR constraint (20) to the no-AN design. This separates the effect of the robust formulation from that of the AN: robust designs are now compared with each other, and nominal designs with each other.

At -10 dB NMSE, the robust no-AN baseline yields an outage probability of $[X]$% with a sensing SNR of $[X]$ dB, compared with $[X]$% and $[X]$ dB for the proposed robust design. The robustness term therefore benefits both designs, and the remaining difference between the two robust designs, $[X]$ dB of sensing SNR, is the cost of the AN, paid in exchange for the reduction of the localization success rate in Fig. 2. We have also removed the statement that attributed the high outage of the nominal no-AN baseline to its higher sensing SNR, since the robust no-AN baseline shows that this outage is removed by the robust formulation alone.

Location in the revised manuscript: Section IV-B, Fig. 4 and its discussion. [Add page and line numbers.]

Revised text:

> *Section IV-B:* "With the same error-averaged constraint, the robust no-AN baseline reduces the outage at -10 dB NMSE to $[X]$% at a sensing SNR of $[X]$ dB. Relative to it, the robust proposed design costs $[X]$ dB of sensing SNR, in exchange for the privacy gain of Fig. 2."

Comments by Reviewer 2

Comment 1:

There is a potential inconsistency in the sensing-cost model. Equation (2) defines the sensing SNR using only the communication/sensing beam matrix $\mathbf{W_k}$ and excludes the AN beam $\mathbf{z_k}$, while the manuscript later states that $s_z[n]$ is known at the sensing receivers and that its echo can be separated from $s_t[n]$ by ideal matched filtering. If $s_z[n]$ is known, its target echo is itself a known probing signal and can, in principle, contribute additional sensing information; if it is deliberately discarded, the reported 3.6 dB sensing-SNR loss is specific to that receiver definition. Can the sensing signal model explicitly state whether the AN echo is exploited, discarded, or treated as interference?

Response:

We thank the reviewer for pointing out this inconsistency. The reviewer is correct that the known AN echo is itself a probing signal, and that the original manuscript did not state how it is handled. The revised sensing model now states explicitly that the AN echo is *separated and discarded*: since $s_z[n]$ is known at the receive APs and independent of the data and sensing symbols, matched filtering separates its echo, and it is neither exploited in (2) nor treated as interference. The residual cross-correlation between independent unit-power symbols after matched filtering decays as $1/N$ in power and is neglected for $N = 128$.

We discard the AN echo in the design deliberately, rather than for convenience. The AN echo from the true target is proportional to $T_k = |\mathbf{a}^H(\phi_k)\mathbf{z}_k|^2$. Including it in the objective would therefore reward AN power in the true direction, which raises the true-direction peak of the beampattern reconstructed by the adversary and works against the fake-peak constraint (9g).

We also agree that the 3.6 dB loss is specific to this receiver definition, and we have qualified it accordingly. If the receivers additionally exploit the AN echo, the sensing SNR becomes $\gamma_t + \sum_k c_k T_k \geq \gamma_t$. The reported γ_t is therefore a lower bound on the achievable sensing SNR, and the reported 3.6 dB loss is an upper bound on the sensing cost of the proposed design. In addition, our response to Reviewer 3, Comment 2, translates this loss into detection and false-alarm probabilities.

Location in the revised manuscript: Section II, sensing-model paragraph; Section III-B, paragraph following (10); Section IV-B, discussion of Fig. 3. [Add page and line numbers.]

Revised text:

> *Section III-B:* "Since $s_z[n]$ is known at the receive APs and independent of the other symbols, matched filtering separates its echo. This echo is excluded from (2) and from the design objective, so that the objective does not reward AN power in the true direction, which would raise T_k. Exploiting it at the receivers would add $\sum_k c_k T_k \geq 0$, so γ_t is a lower bound on the achievable sensing SNR."
>
> *Section IV-B:* "Since the AN echo is discarded in (2), this loss is an upper bound on the sensing cost of the proposed design."

Comment 2:

The entire threat model relies on the assumption that the adversarial user can separate its received superposition into per-AP raw components and then reconstruct $\hat{\mathbf{x}}_k[n]$ for each AP. However, the actual observation is $\mathbf{y}_a[n] = \sum_k \mathbf{H}a, k\mathbf{x}_k[n] + \mathbf{n}_a[n]$, while each $\mathbf{x}_k[n] \in \mathbb{C}^{M_{\text{tx}}}$ is unknown; with $N_{\text{tx}}M_{\text{tx}}$ transmit dimensions but only M_{UE} received dimensions per snapshot, this decomposition is generally underdetermined. What identifiability conditions make the assumed EM reconstruction of each AP waveform possible, particularly for the default setting $N_{\text{tx}} = 8$, $M_{\text{tx}} = 16$, and $M_{\text{UE}} = 16$?

Response:

We thank the reviewer for this insightful comment and fully agree with the dimension count. Since the APs transmit common symbols, $\mathbf{y}_a[n]$ offers $M_{\text{UE}} = 16$ observation dimensions per snapshot, against $N_{\text{tx}}M_{\text{tx}} = 128$ per-AP transmit dimensions in the default setting. The decomposition of $\mathbf{y}_a[n]$ into per-AP components is therefore underdetermined, and the original manuscript did not state clearly that this step is an assumption rather than a consequence of the signal model. We have revised the threat model to separate the attack into two stages and to state the identifiability status of each.

(i) Per-AP separation (assumed). The separation of $\mathbf{y}_a[n]$ into per-AP components is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [14], we now state explicitly that we adopt a genie-aided strong-adversary model, in which the per-AP observations $\widetilde{\mathbf{y}}_{a,k}[n]$ are available to the adversary. The genie removes the inter-AP superposition, which is the main obstacle for a practical adversary, so the privacy of the proposed design is assessed against an adversary that is granted a capability that a practical adversary does not have in general. Since the network cannot verify the separation capability of a served user, we consider it appropriate that the defense does not rely on this capability being limited. Partial separation may become available in practice, e.g., when AP-specific transmissions are resolvable, or under wideband operation with resolvable inter-AP delays.

(ii) Per-AP reconstruction (identifiable under a stated condition). Given $\widetilde{\mathbf{y}}_{a,k}[n] = \widehat{\mathbf{H}}^{\text{adv}}_{a,k}\widetilde{\mathbf{x}}_k[n] + \widetilde{\mathbf{n}}^{\text{adv}}_{a,k}[n]$, the recovery of $\widetilde{\mathbf{x}}_k[n]$ is a linear inverse problem with M_{UE} equations and M_{tx} unknowns. It is identifiable when $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$ has full column rank, which holds with probability one under Rayleigh fading whenever $M_{\text{UE}} \geq M_{\text{tx}}$. The default setting satisfies this condition with $M_{\text{UE}} = M_{\text{tx}} = 16$. Moreover, since the beampattern in (3) depends only on the covariance $\mathbf{R}_{\widetilde{\mathbf{x}}_k}$, whose rank is at most $N_{\text{UE}} + 1$ (the other users' streams, the sensing stream, and the AN stream), a weaker sufficient condition is that $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$ is injective on the column space of $[\mathbf{W}_{-a,k}, \mathbf{z}_k]$. If $M_{\text{UE}} < M_{\text{tx}}$, the adversary recovers only the projection of $\mathbf{R}_{\widetilde{\mathbf{x}}_k}$ onto the row space of $\widehat{\mathbf{H}}^{\text{adv}}_{a,k}$, and the reconstructed beampattern is distorted, which further favors the defender. The EM routine of [14] jointly refines the channel and signal estimates within this per-AP linear model. It does not resolve the inter-AP superposition, which is removed by the genie.

We have also rewritten the sentence explaining why the attack operates on raw samples, since in the original version it could be read as implying that raw samples alone suffice for per-AP recovery. In addition, Section IV now states that the simulated adversary is given accurate knowledge of $\mathbf{H}_{a,k}$, and that the reported success rates therefore correspond to a conservative attack model.

Location in the revised manuscript: Section II, threat-model paragraph; Section IV-A, simulation setup; Section IV-B, first paragraph. [Add page and line numbers.]

Revised text:

> *Section II:* "One served user, indexed by a, is treated as an internal adversary. It knows its own pilots, data symbols, combiner $\mathbf{u}_a$, and the transmit-AP positions, and can form its own estimate $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$ of its channel $\mathbf{H}_{a,k}$. Since all APs transmit common symbols, $\mathbf{y}_a[n]$ offers only M_{UE} dimensions for $N_{\mathrm{tx}}M_{\mathrm{tx}}$ per-AP transmit dimensions, so its per-AP decomposition is not identifiable from $\mathbf{y}_a[n]$ alone in general. Following [14], we therefore adopt a genie-aided strong-adversary model in which the per-AP components of the received samples are available to the adversary, as may occur when AP-specific transmissions are resolvable in time, frequency, or delay. Since the network cannot verify the receiver capability of a served user, this conservative model avoids relying on the adversary being unable to separate them. The attack operates on raw multi-antenna samples, since applying $\mathbf{u}_a$ would collapse each snapshot into a scalar. After decoding its own data symbol $s_a[n]$, the adversary subtracts its data component using its channel estimate $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$, which leaves $\widetilde{\mathbf{y}}_{a,k}[n] = \widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}\widetilde{\mathbf{x}}_k[n] + \widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ with the non-self transmit component $\widetilde{\mathbf{x}}_k[n] = \sum_{j\neq a}\mathbf{w}_{j,k}s_j[n] + \mathbf{w}_{t,k}s_t[n]$, and $\widetilde{\mathbf{n}}_{a,k}^{\mathrm{adv}}[n]$ collecting the receiver noise and the residual caused by adversary-side channel-estimation errors. It then applies an expectation-maximization routine to obtain $\widehat{\mathbf{x}}_k[n] \approx \widetilde{\mathbf{x}}_k[n]$, which is identifiable when $\widehat{\mathbf{H}}_{a,k}^{\mathrm{adv}}$ has full column rank, i.e., with probability one for $M_{\mathrm{UE}} \geq M_{\mathrm{tx}}$, and forms the AP-dependent beampattern from the sample covariance over the available snapshots."

Comment 3:

The literature review should be improved with recent works on multi-static ISAC:
R1— A. Chowdhury, S. S. Thoota and E. G. Larsson, "Joint Sensing and Bi-Directional Communication With Dynamic TDD Enabled Cell-Free MIMO," in IEEE Transactions on Wireless Communications, vol. 25, pp. 2420–2434, 2026, doi: 10.1109/TWC.2025.3596622.
R2—- X. Sun et al., "Performance Analysis and Optimization for Network-Assisted Full-Duplex Cell-Free Multi-Static ISAC Systems," in IEEE Transactions on Green Communications and Networking, vol. 10, pp. 272–288, 2026, doi: 10.1109/TGCN.2025.3575837.

Response:

We thank the reviewer for pointing out these recent works. Both are now discussed and cited in the introduction as [4] and [5], as examples of multi-static cell-free ISAC with dynamic time-division duplexing and with network-assisted full-duplex operation, respectively.

Location in the revised manuscript: Section I, first paragraph. [Add page and line numbers.]

Revised text:

> *Section I:* "Multi-static cell-free ISAC has further been studied with dynamic time-division duplexing for joint sensing and bidirectional communication [4] and with network-assisted full-duplex operation [5]."

Comment 4:

Network-wide effective-channel nulling in (6) relies on coherent cancellation of the AN contributions transmitted by geographically distributed APs, i.e., $\sum_k \widehat{\mathbf{g}}_{i,k}^H \mathbf{z}_k = 0$. This cancellation is sensitive not only to channel-estimation errors but also to residual inter-AP carrier-phase, timing, and frequency synchronization errors. Since the imperfect-CSI extension in (17)–(20) accounts only for isotropic channel-estimation errors, how robust is the central receive-processing asymmetry when the distributed AP phases are not perfectly synchronized?

Response:

We thank the reviewer for raising this important practical issue. We agree that the effective-channel null relies on coherent cancellation across the APs and that (17)–(20) do not cover synchronization errors. We have added an analysis and a numerical evaluation of this effect.

What is affected. A residual phase offset ψ_k at AP k rotates its entire transmit signal. This affects only the coherent cancellation at the legitimate combiner outputs. The sensing SNR (2) is unaffected, since the receive APs combine the echoes of different transmit APs non-coherently, as clarified in our responses to Reviewer 1, Comment 4, and Reviewer 3, Comment 1. The privacy mechanism is unaffected as well, since the adversary processes each AP's component separately, and a per-AP phase rotation leaves each covariance $\mathbf{R}_{\widehat{\mathbf{x}}_k}$ unchanged.

Residual AN. With $c_{i,k} = \widehat{\mathbf{g}}_{i,k}^H \mathbf{z}_k$ and independent phase errors $\psi_k \sim \mathcal{N}(0, \sigma_\psi^2)$,

$$\mathbb{E}\left\{\left|\textstyle\sum_k e^{\jmath\psi_k} c_{i,k}\right|^2\right\} = e^{-\sigma_\psi^2}\left|\textstyle\sum_k c_{i,k}\right|^2 + \left(1 - e^{-\sigma_\psi^2}\right)\textstyle\sum_k |c_{i,k}|^2 = \left(1 - e^{-\sigma_\psi^2}\right)\textstyle\sum_k |c_{i,k}|^2,$$

where the last step uses (6). The residual AN power therefore grows approximately as $\sigma_\psi^2 \sum_k |c_{i,k}|^2$, i.e., quadratically in the phase standard deviation and in proportion to the per-AP AN contributions that the null cancels. Since this term is a convex quadratic form in the AN beams, it can be added to the robust formulation in the same way as the CSI-error floor L_i^{csi}. The same expression applies to the data and sensing beams, whose coherent gains are scaled by $e^{-\sigma_\psi^2}$.

Numerical evaluation. For phase standard deviations of 1, 2, 5, 10, and 20°, the residual AN power at the combiner outputs is $[X]$, $[X]$, $[X]$, $[X]$, and $[X]$ dB relative to the noise, in agreement with the closed form to within $[X]$ dB. The outage probability of the proposed design is $[X]$%, $[X]$%, $[X]$%, $[X]$%, and $[X]$%, compared with $[X]$%, $[X]$%, $[X]$%, $[X]$%, and $[X]$% for the no-AN baseline. The receive-processing asymmetry is thus preserved for phase errors up to about $[X]$°. Residual phase errors of this kind are what over-the-air synchronization and reciprocity-calibration protocols for distributed MIMO address [16], and the analysis above specifies the accuracy they must achieve for the proposed design. Beyond this accuracy, the robust extension above is required.

Location in the revised manuscript: Section III-C, new remark after (20); Section IV-B. [Add page and line numbers.]

Revised text:

> *Section III-C:* "Residual inter-AP phase offsets $\psi_k \sim \mathcal{N}(0, \sigma_\psi^2)$ break the coherent cancellation in (6) and leave the AN power $(1 - e^{-\sigma_\psi^2}) \sum_k |\widehat{\mathbf{g}}_{i,k}^H \mathbf{z}_k|^2$ at the output of combiner i. Since this term is a convex quadratic form, it can be added to $\overline{\mathcal{I}}_i$ in the

same way as L_i^{csi}. Such phase offsets are the target of over-the-air synchronization protocols for distributed MIMO [16]. The sensing SNR (2) and the attack, which process the transmit APs separately, are unaffected."

Section IV-B: "With phase standard deviations up to $[X]°$, the residual AN stays $[X]$ dB below the noise and the outage of the proposed design remains below $[X]$%."

Comment 5:

The privacy evaluation assumes that the adversary always selects the dominant peak of each reconstructed beampattern and intersects those directions. An adversary aware of the fake-peak mechanism could instead retain multiple significant peaks, form all geometrically consistent multi-AP intersections, or explicitly test the hypothesis that one coherent set of peaks is artificial. What property of the proposed AN makes the true target location statistically unidentifiable under such a multi-hypothesis adversary? As currently formulated, the reported localization-success reduction appears to certify robustness against one specific reconstruction rule rather than sensing-location privacy in a broader sense.

Response:

We thank the reviewer for this insightful comment and agree with its conclusion. The reported reduction certifies robustness against the dominant-peak reconstruction rule of [14], and we have revised the claims of the manuscript accordingly. Below, we identify the property that governs a multi-hypothesis adversary, state what the proposed design can and cannot guarantee against it, and report its performance against such an adversary.

What the AN can and cannot do. The AN symbol $s_z[n]$ is independent of the sensing symbol $s_t[n]$, so the AN adds terms to the reconstructed covariance but cannot cancel the true-direction component $\mathbf{a}^H(\phi_k)\mathbf{W}_{-a,k}$, which the legitimate sensing requires. Against an adversary that retains several peaks, the AN therefore creates a competing hypothesis rather than erasing the true one. Privacy against such an adversary then means that the adversary cannot tell the true hypothesis from the artificial one.

The governing property: statistical symmetry of the hypotheses. The adversary knows neither $s_t[n]$ nor $s_z[n]$. Both are zero-mean, unit-variance, mutually independent, and independent of the adversary's own data, so the reconstructed covariance contains the true-direction and fake-direction components as rank-one terms driven by unknown streams with identical statistics. By construction, the fake directions of the APs in $\mathcal{K}_{\text{res}}$ are geometrically consistent with the common virtual location $\mathbf{p}_{\text{f}}$, just as the true directions are consistent with $\mathbf{p}_t$. This symmetry, however, also requires the fake hypothesis to be supported by the same APs as the true one. An AP whose true and fake directions are unresolvable supports only the true hypothesis, so that a vote-counting adversary favors the target unless $\mathcal{K}_{\text{res}}$ contains all transmit APs. In the revised simulations, we therefore choose $\mathbf{p}_{\text{f}} = (175, 35)$ m, for which all eight transmit APs satisfy the resolvability condition of our response to Reviewer 1, Comment 2. With full support, the two hypotheses differ only in their peak powers, which are set by ρ in (9g). If the two powers are matched, the peak powers carry no information on which hypothesis is artificial, and an adversary that relies on the reconstructed beampatterns faces two hypotheses that are, up to sidelobe details, symmetric. It can then identify the target with probability at most one half of the probability that the target is among the hypotheses, and with L virtual locations this bound becomes $1/(L+1)$. We note that the symmetry is not complete: the true-direction response collects several unknown streams, whereas the

fake-direction response is generated by the single AN stream, so an adversary that analyzes the eigen-structure of $\mathbf{R}_{\widehat{\mathbf{x}}_k}$ could in principle exploit this difference. Spreading the AN over several streams to remove it is noted as an extension. Conversely, with $\rho > 1$, the fake peak is systematically the stronger one: this maximizes the deception of the dominant-peak attack, but an adversary that knows the mechanism can exploit the power asymmetry.

Design implication. The parameter ρ therefore trades the deception of a dominant-peak adversary against the indistinguishability of the hypotheses for a mechanism-aware adversary. To make the matched-power operating point available, the design now optionally caps the fake-direction response, $F_k \leq \kappa(\rho\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\|_F^2 + T_k)$ with $\kappa \geq 1$, which in the CCCP update is again a convex constraint when its right-hand side is replaced by its affine lower bound. With $\rho \approx 1$, the two peak powers are then matched to within the factor κ. In addition, $\mathbf{p}_\mathrm{f}$ should be defined relative to the target position, e.g., by a fixed offset, so that an adversary observing several frames cannot single out the artificial hypothesis as the one that does not follow the target's motion.

Evaluation. The revised manuscript evaluates a multi-hypothesis adversary. It retains the two strongest distinct peaks of each $\widehat{B}_k(\theta)$, casts one direction line per peak, forms the two most-voted, geometrically separated location hypotheses, and resolves them by the vote count, at random, or, knowing that the fake peak is designed to be the stronger one, by choosing the hypothesis supported by the weaker peaks. The success of an adversary allowed two guesses bounds every rule that picks one of the two hypotheses. For the proposed design, the success rates of the vote, random, and mechanism-aware rules and the two-guess success are [X]%, [X]%, [X]%, and [X]% at $\rho = [X]$ ($\eta \approx 0.50$), and [X]%, [X]%, [X]%, and [X]% at $\rho \approx 1$, compared with [X]%, [X]%, [X]%, and [X]% without AN.

Revised claims. We have revised the abstract, Section IV, and the conclusion to state that the reported reduction of the localization success rate holds for the dominant-peak attack, and that against a multi-hypothesis adversary the design provides ambiguity between the true and virtual locations, whose level is controlled by ρ. A design with several virtual locations, which would lower the bound to $1/(L+1)$, is noted as an extension.

Location in the revised manuscript: Abstract; Section II, threat model; Section III-B, remark after (10); Section IV-B; Section V. [Add page and line numbers.]

Revised text:

> *Section III-B, remark after (10):* "Since $s_t[n]$ and $s_z[n]$ are both unknown to the adversary and statistically identical, and since $\mathbf{p}_\mathrm{f}$ is chosen such that $\mathcal{K}_\mathrm{res}$ contains all transmit APs, the true and fake hypotheses are supported by the same APs and differ only in their peak powers. With $\rho \approx 1$ and the fake-direction response capped as $F_k \leq \kappa(\rho\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\|_F^2 + T_k)$, the two powers are matched, and a beampattern-based adversary can then identify the target with probability at most one half of the probability that it is among the hypotheses. A larger ρ instead maximizes the deception of the dominant-peak attack."
>
> *Abstract and Section V:* "...the proposed design lowers the success rate of the dominant-peak reconstruction attack from 35.8% to [X]% ... Against a multi-hypothesis adversary, it creates ambiguity between the true and virtual locations, whose level is controlled by ρ."

Comment 6:

Minor corrections. In (7), the stated null-space dimensions d_{phys} and d_{eff} hold under the implicit assumption that the corresponding stacked channel matrices have full possible row rank; otherwise these should be presented as generic-dimension expressions rather than unconditional equalities. In the imperfect-CSI discussion, "independent of $\hat{\mathbf{H}}$ and across links, with i.i.d. entries" should distinguish independence across links from identical variances, since $\beta_{e,i,k}$ is link dependent. It would also be useful to state explicitly the feasibility condition rank($\mathbf{Q}_i$) [. . .]

Response:

We thank the reviewer for these careful corrections, which we have addressed as follows.

- *Null-space dimensions in (7).* We agree. Since the dimension of a null space equals the number of columns minus the rank, and the rank cannot exceed the number of rows, (7) now states the lower bounds $d_{\text{phys}} \geq N_{\text{tx}}[M_{\text{tx}} - N_{\text{UE}}M_{\text{UE}}]^+$ and $d_{\text{eff}} \geq [N_{\text{tx}}M_{\text{tx}} - N_{\text{UE}}]^+$, with equality when the stacked matrices have full row rank, which holds with probability one for i.i.d. Rayleigh fading and generic combiners.
- *Independence versus identical variances.* We agree and have revised the sentence to state that $\mathbf{E}_{i,k}$ is independent of $\widehat{\mathcal{H}}$ and independent across links, and that its entries are i.i.d. within each link with the link-dependent variance $\widetilde{\beta}_{i,k}$.
- *Feasibility of the combiner update.* The constrained MMSE update (15) requires the set $\{\mathbf{u}_i : \mathbf{Q}_i^H \mathbf{u}_i = \mathbf{0}, \|\mathbf{u}_i\| = 1\}$ to be nonempty, i.e., rank($\mathbf{Q}_i$) $< M_{\text{UE}}$, and the vector $\widetilde{\mathbf{u}}_i$ to be nonzero. Since $\mathbf{R}_i^{-1}$ is invertible, $\widetilde{\mathbf{u}}_i \neq \mathbf{0}$ holds if and only if $\mathbf{q}_{i,i} \notin \text{span}(\mathbf{Q}_i)$. As $\mathbf{Q}_i$ has two columns, the first condition holds whenever $M_{\text{UE}} > 2$, which includes the default $M_{\text{UE}} = 16$. Both conditions are now stated explicitly after (16).

Location in the revised manuscript: Section III-A, (7); Section III-B, after (16); Section III-C, first paragraph. [Add page and line numbers.]

Revised text:

Section III-A, (7):

$$d_{\text{phys}} \geq N_{\text{tx}}[M_{\text{tx}} - N_{\text{UE}}M_{\text{UE}}]^+, \quad d_{\text{eff}} \geq [N_{\text{tx}}M_{\text{tx}} - N_{\text{UE}}]^+,$$

"with equality when the stacked channel matrices have full row rank, which holds with probability one under i.i.d. Rayleigh fading."

Section III-B, after (16): "The update is well defined when rank($\mathbf{Q}_i$) $< M_{\text{UE}}$, which holds for $M_{\text{UE}} > 2$, and $\mathbf{q}_{i,i} \notin \text{span}(\mathbf{Q}_i)$, which ensures $\widetilde{\mathbf{u}}_i \neq \mathbf{0}$."

Section III-C: "Under the LMMSE model, $\mathbf{E}_{i,k}$ is independent of $\widehat{\mathcal{H}}$ and independent across links, and its entries are i.i.d. $\mathcal{CN}(0, \widetilde{\beta}_{i,k})$ within each link, with the link-dependent variance $\widetilde{\beta}_{i,k} = \ldots$"

Comments by Reviewer 3

Comment 1:

Most of all, the legitimate sensing operation is not clearly described. For example, although multiple RX APs are considered, the operation at each RX AP (e.g., receive beamforming) and the combining between APs (coherent/non-coherent, central/distributed) are not described. It is difficult to grasp how the sensing SINR in Eq.2 is obtained. Also, it is not clear who experiences the sensing SINR. Is it an individual RX AP, or the system as a whole?

Response:

We thank the reviewer for this comment and agree that the legitimate sensing operation was not described in sufficient detail. The revised Section II now describes the processing chain as follows.

1. *Operation at each RX AP.* Each receive AP employs M_{rx} antennas and steers its receive beam toward the target. The resulting receive array gain is absorbed, together with the target RCS, in the scattering power $\xi_{r,k}^2$. Each receive AP then matched-filters its received samples with the known transmitted symbols. The data symbols are available at the central processor and therefore act as additional probing signals, while the echo of the known AN symbol is separated by matched filtering and discarded.

2. *Combining between RX APs.* The receive APs forward their matched-filter outputs to the central processor, which fuses them non-coherently. Non-coherent fusion avoids phase synchronization among the receive APs.

3. *How (2) is obtained.* At receive AP r, the echo is the superposition

$$\sum_{k=1}^{N_{\mathrm{tx}}} \alpha_{r,k}\sqrt{g_{r,k}}\,\mathbf{a}^H(\phi_k)\mathbf{W}_k\mathbf{s}[n],$$

   where $\mathbf{s}[n]$ collects the unit-power symbols and $\alpha_{r,k}$ is the bistatic echo coefficient with $\mathbb{E}\{|\alpha_{r,k}|^2\} = \xi_{r,k}^2$. Since the echo phases are not known to within the carrier wavelength, they are modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases and the symbols gives the per-snapshot SNR at receive AP r,

$$\gamma_r = \frac{1}{\sigma_{\mathrm{rx}}^2}\sum_{k=1}^{N_{\mathrm{tx}}} \xi_{r,k}^2 g_{r,k}\left\|\mathbf{a}^H(\phi_k)\mathbf{W}_k\right\|_F^2,$$

   and (2) is its average $\gamma_t = \frac{1}{N_{\mathrm{rx}}}\sum_r \gamma_r$.

4. *Who experiences the SNR.* γ_t is the average per-receive-AP SNR. The SNR of the fused statistic at the central processor is $N_{\mathrm{rx}}\gamma_t$. Since the two differ by a constant factor, they are maximized by the same beams, and the design is unaffected by this choice. All reported sensing SNRs are average per-receive-AP values, which is now stated in Section IV.

The detector applied to the fused statistic is described in our response to Comment 2.

Location in the revised manuscript: Section II, sensing-model paragraph; Section IV-A, simulation setup. [Add page and line numbers.]

Revised text:

> *Section II:* "Each receive AP employs M_{rx} antennas, steers its receive beam toward the target, and matched-filters its received samples with the known transmitted symbols, where the data symbols, being available at the central processor, act as additional probing signals. The bistatic echo phases are not known to the network to within the carrier wavelength and vary with the aspect angle, and are therefore modeled as independent and uniformly distributed across transmit–receive AP pairs. Averaging over these phases removes the cross terms between the echoes of different transmit APs. With $\mathbf{W}_k \triangleq [\mathbf{w}_{1,k}, \ldots, \mathbf{w}_{N_{\mathrm{UE}},k}, \mathbf{w}_{t,k}]$, the per-snapshot sensing SNR, averaged over the echo phases and the N_{rx} receive APs, is [(2), unchanged], where ..., $\xi_{r,k}^2$ is the scattering power absorbing the target radar cross section (RCS) and the receive array gain, and σ_{rx}^2 is the receive noise power. The receive APs forward their matched-filter outputs to the central processor, which fuses them non-coherently, so the fused SNR $N_{\mathrm{rx}}\gamma_t$ is maximized by the same beams as γ_t."

Comment 2:

Furthermore, the authors have not described any detector or detection threshold. Therefore, "x dB sensing SINR loss" doesn't represent anything. How is it translated into the detection and false-alarm probabilities, which eventually matter? Without this, it is very difficult to judge whether the proposed scheme is practically useful.

Response:

We thank the reviewer for this important comment and agree that the SNR loss should be related to detection performance. The revised manuscript now specifies the detector and reports the corresponding detection probabilities.

Detector. Each receive AP matched-filters its samples with the $N_{\mathrm{s}} = N_{\mathrm{UE}} + 1$ known data and sensing streams. With unknown echo amplitudes and phases, the generalized likelihood ratio test sums the matched-filter output energies over the N_{s} streams and the N_{rx} receive APs, and compares the sum with a threshold τ. After normalization, the statistic follows a central χ^2 distribution with $2N_{\mathrm{rx}}N_{\mathrm{s}}$ degrees of freedom under the null hypothesis, so τ is set by $\Pr\{\chi^2_{2N_{\mathrm{rx}}N_{\mathrm{s}}} > \tau\} = P_{\mathrm{fa}}$. Under the target hypothesis, it follows a noncentral χ^2 distribution with noncentrality $2N\sum_r \gamma_r$, which gives

$$P_{\mathrm{d}} = \mathbb{E}\Big\{Q_{N_{\mathrm{rx}}N_{\mathrm{s}}}\Big(\sqrt{2N\textstyle\sum_r \gamma_r}, \sqrt{\tau}\Big)\Big\},$$

where $Q_M(\cdot,\cdot)$ is the generalized Marcum Q-function of order M. Consistent with the random echo phases of the sensing model, the per-receive-AP SNRs γ_r are taken as independent and exponentially distributed with mean γ_t, and the expectation is over this fluctuation. Among fluctuation models with the same mean, this is a conservative choice in the high-P_{d} regime.

Detection performance. For $P_{\mathrm{fa}} = 10^{-6}$, $N_{\mathrm{rx}} = 4$, and $N_{\mathrm{s}} = 5$, $P_{\mathrm{d}} = 0.9$ requires $\gamma_t \geq -8.5$ dB with integration over the $N = 128$ snapshots. The proposed design at $\eta \approx 0.50$ ($\gamma_t = 11.1$ dB) therefore retains a margin of 19.6 dB, compared with 23.2 dB for the no-AN reference ($\gamma_t = 14.7$ dB), and both designs achieve $P_{\mathrm{d}} > 0.999$. Even with the shortest

integration that resolves all streams, $N = N_s = 5$, the proposed design achieves $P_d = 0.998$, compared with 0.9999 for the no-AN reference. The 3.6 dB loss therefore reduces the detection margin rather than the detection probability in the default setting.

Practical interpretation. Since the two-way path gain $g_{r,k}$ scales with the fourth power of distance, a 3.6 dB loss corresponds to an approximately 19% reduction of the bistatic distances that can be supported at fixed (P_d, P_{fa}). Moreover, as stated in our response to Reviewer 2, Comment 1, the reported loss is an upper bound, since the known AN echo is discarded. Given the length limit of a correspondence, we report these values in the text of Section IV-B rather than in an additional figure.

Location in the revised manuscript: Section IV-B, discussion of Fig. 3. [Add page and line numbers.]

Revised text:

> *Section IV-B:* "To relate these values to detection performance, we consider the generalized likelihood ratio test at the central processor, which sums the matched-filter output energies over the $N_{UE} + 1$ known streams and the N_{rx} receive APs, with the threshold set for $P_{fa} = 10^{-6}$. With per-receive-AP SNRs exponentially distributed around γ_t due to the random echo phases, $P_d = 0.9$ requires $\gamma_t \geq -8.5$ dB over $N = 128$ snapshots, so the proposed design at $\eta \approx 0.50$ retains a 19.6 dB margin, compared with 23.2 dB without AN, and both achieve $P_d > 0.999$. The 3.6 dB loss thus reduces the detection margin rather than the detection probability, and corresponds to an approximately 19% reduction of the supported bistatic distances at fixed (P_d, P_{fa})."

Comment 3:

All comparisons are between different AN schemes. Considering that AN spends a significant portion of power, another baseline to compare with would be a no-AN alternative that allocates the corresponding power to communication while keeping comparable sensing performance. The authors should remember that it is an ISAC system. It is questionable whether treating communication merely as a minimum requirement while devoting more than half of the transmit power to creating a fake peak is a reasonable design choice for an ISAC system. The authors should elaborate further on the tradeoff between sensing, communication, and security.

Response:

We thank the reviewer for this important comment. We agree that the AN power has an opportunity cost in communication, and that the original manuscript did not quantify it. We have added the suggested baseline and a discussion of the three-way tradeoff.

No-AN alternative with the AN power spent on communication. For each channel realization, we give the no-AN design the same sensing SNR as the proposed design at $\eta \approx 0.50$ and find, by bisection, the largest common SINR target it can still meet. This target is $[X]$ dB, i.e., $[X]$ dB above the 10 dB requirement, and the corresponding sum rate is $[X]$ bit/s/Hz, compared with $[X]$ bit/s/Hz for the proposed design. This rate gap is the communication cost of the privacy provided at this operating point.

Three-way tradeoff. Table R1 summarizes, for several AN-power ratios, the localization success rate, the sensing SNR, and the rate gap with respect to the no-AN alternative. The

operator selects the operating point through ρ according to the required privacy level. For example, a smaller AN-power ratio of $\eta \approx [X]$ already reduces the localization success rate to $[X]\%$ at a rate gap of $[X]$ bit/s/Hz. Moreover, the AN power is not entirely lost for sensing: as discussed in our response to Reviewer 2, Comment 1, exploiting the known AN echo recovers $[X]$ dB of sensing SNR.

Design rationale. We treat communication as a guaranteed QoS constraint because the privacy requirement is imposed in addition to an ISAC service that must keep serving its users. The new baseline makes explicit what this guarantee costs relative to a design that would spend the AN power on communication, and we have added this discussion to Section IV.

Table R1. Tradeoff between privacy, sensing, and communication.

η	Localization success (%)	Sensing SNR (dB)	Rate gap (bit/s/Hz)
0 (no AN)	$[X]$	$[X]$	0
$[X]$	$[X]$	$[X]$	$[X]$
$[X]$	$[X]$	$[X]$	$[X]$
≈ 0.50	$[X]$	$[X]$	$[X]$

Location in the revised manuscript: Section IV-B, new paragraph after the discussion of Fig. 3. [Add page and line numbers.]

Revised text:

> *Section IV-B:* "To quantify the communication cost of privacy, we give the no-AN design the sensing SNR of the proposed design at $\eta \approx 0.50$ and maximize its common SINR target. It then supports $[X]$ dB instead of 10 dB, i.e., a sum rate of $[X]$ instead of $[X]$ bit/s/Hz. This rate gap, together with the sensing-SNR loss of Fig. 3, is the price of reducing the localization success rate to $[X]\%$, and the operator trades it through ρ."

Comment 4:

In the experiment, the no-AN localization success rate is 36%, which is already low. What if the adversary is in a more favorable situation? Is AN still effective?

Response:

We thank the reviewer for this question. The no-AN success rate of about 36% reflects a strict success criterion, namely that the estimate falls in the exact 10 m $\times$ 10 m cell of the target, together with the front-back ambiguity of the uniform linear arrays and the reconstruction errors from a finite number of snapshots. Several assumptions already favor the adversary: it is genie-aided in the per-AP separation, as discussed in our responses to Reviewer 1, Comment 1, and Reviewer 2, Comment 2, and the simulated adversary knows its channels accurately.

To examine an even more favorable situation, we have evaluated the attack with coarser search cells, which make success easier. With 20 m and 50 m cells, the no-AN success rate rises to $[X]\%$ and $[X]\%$, while the proposed design at $\eta \approx 0.50$ keeps it at $[X]\%$ and $[X]\%$, respectively. In addition, an adversary allowed two guesses, as in our response to Reviewer 2, Comment 5, succeeds in $[X]\%$ of the cases without AN and $[X]\%$ with the proposed design. The AN

therefore remains effective when the adversary is favored, since it moves the estimate to a different location rather than merely degrading its accuracy.

Location in the revised manuscript: Section IV-B, discussion of Fig. 2. [Add page and line numbers.]

Revised text:

> *Section IV-B:* "For a more favorable adversary, coarser search cells of 20 and 50 m raise the no-AN success rate to [X]% and [X]%, whereas the proposed design keeps it at [X]% and [X]%."

Comment 5:

Minor comments.

- RCS is not specified in the experiments.
- Eq. 2 is not really an SINR. No interference there.
- TheRX-AP antenna number is missing.

Response:

We thank the reviewer for these careful observations, which we have addressed as follows.

- *RCS.* The target RCS used in the simulations, $\sigma_{\mathrm{RCS}} = 10$ m^2 (10 dBsm), is now listed in Table I, and Section II states that $\xi_{r,k}^2$ absorbs the RCS and the receive array gain.
- *SNR versus SINR.* We agree. Since the data symbols are known probing signals and the AN echo is separated by matched filtering, (2) contains no interference term and is an SNR. The revised manuscript uses the term "sensing SNR" consistently and states that the AN echo is separated and discarded.
- *RX-AP antenna number.* The number of receive-AP antennas is now defined in Section II and listed in Table I. The simulations use $M_{\mathrm{rx}} = 1$, i.e., no receive array gain in $\xi_{r,k}^2$; with M_{rx} antennas steered toward the target, all sensing SNRs would increase by $10 \log_{10} M_{\mathrm{rx}}$ dB, which does not change the comparison between the designs.

Location in the revised manuscript: Section II, sensing-model paragraph; Table I. [Add page and line numbers.]

Revised text:

> *Table I (added rows):* "Target RCS σ_{RCS}: 10 m^2; RX-AP antennas M_{rx}: 1."